Master's Comprehensive Examination

Part I | Theory (90 minutes) | December 4, 1999

Instructions: Answer all four questions. Open book and open notes.

1. Let X be a random variable whose density is

$$f_X(x) = \begin{cases} cx^3 & \text{if } 0 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find c.
(b) Find $P\{\frac{1}{4} \leq X < \frac{3}{4}\}$.
(c) Find the cdf $F_X(x)$ of X.

2. Let X_1, X_2 be a random sample of size 2 from a Poisson distribution with parameter λ. Test $H_0 : \lambda = 1$ vs $H_1 : \lambda > 1$. Find the UMP test of size $\alpha = .0526$. (Note that $(X_1 + X_2) \sim P(2\lambda)$.)

3. On a three-part exam, the probability is 1/2 of correctly answering the first part. The probability of correctly answering the second part, given that the first part is correct, is 3/4; the probability of being correct on the second part, given that the first part is incorrect, is 1/4. Finally, the probability of correctly answering part 3 is

2/5 given both previous parts are missed;

1/5 given part 2 is wrong, part 1 is correct;

4/5 given part 1 is wrong, part 2 is correct;

3/5 given part 1 is correct, part 2 is correct.

Find the probability that

(a) All three are wrong.

(b) Only part 2 is correct.

(c) Exactly two parts are correct.

4. Let X_1, X_2 be independent normal with mean μ and variance 2.

(a) Find a sufficient statistic.

(b) Is $e^{X_1+X_2}$ a sufficient statistic? Why or why not?

(c) Find the MSE of the estimator $T = \frac{X_1+X_2}{2} + 3$.